CASE STUDY · AI AUTOMATION

RoofRadar AI: instant lead response and booking for roofing companies

An AI receptionist that replies to every lead in seconds, qualifies it and books the inspection into the real calendar.

ROLE	DEMO CLIENT	STACK
AI Automation Engineer (solo build): Fatima Ali	PrimeHome Roofing, NY/NJ	n8n · OpenAI · Twilio · Google Calendar · Postgres/Supabase · Slack · HubSpot

The problem

Roof repairs are high-value jobs, and a homeowner with a leak contacts two or three roofers and books whoever answers first. The people who should answer are on roofs, driving or quoting jobs, so leads wait.

- **Slow first reply.** Web and Facebook leads arrive in the evening; the office replies the next morning, after a competitor already has.
- **Missed calls stay missed.** Voicemail sends the homeowner to the next roofer on the list.
- **Booking is back-and-forth**, and two people working one calendar cause double bookings.
- **No follow-up** on quiet leads, no-shows and open quotes.
- **Generic AI chatbots are risky:** they invent prices, promise times that aren't free and keep texting people who replied STOP.

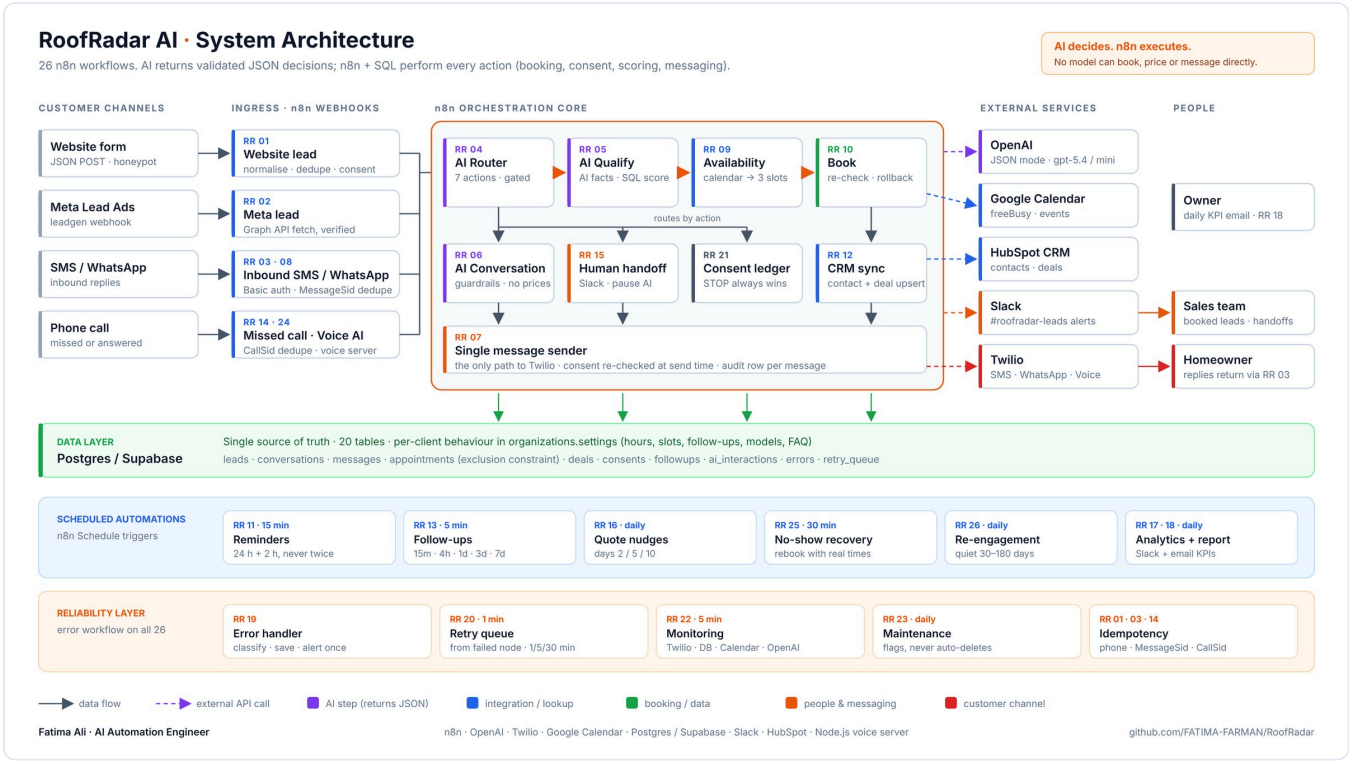
The solution

RoofRadar AI is a set of 26 n8n workflows that handle the lead from first contact to a booked inspection. Every lead from the website, Facebook, SMS, WhatsApp or a missed call gets a personal reply within seconds. OpenAI works out the problem, urgency and location; the system offers three real free times from Google Calendar; the homeowner replies "2" and the inspection is booked, confirmed by text and posted to the sales team on Slack. Price questions, complaints and emergencies go straight to a person, and the AI pauses for that lead.

Core rule: the AI decides, n8n executes. Models only return validated JSON decisions. Availability, bookings, lead scores, consent checks, CRM writes and every outbound message are deterministic n8n and SQL, so the AI can't invent a price or a time.

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from web form to first text in testing	n8n workflows, one shared database	test executions / failures	double bookings: enforced by the database

How it works



System architecture: lead channels → n8n core → AI decisions → booking, messaging and operations.

#	WHAT THE HOMEOWNER SEES	WHAT RUNS
1	Submits the PrimeHome form at 9:12 PM: roof leak, Newark NJ	Lead saved, deduplicated, consent recorded (224 ms)
2	"Hi John, thanks for contacting PrimeHome Roofing about your roof leak..."	AI writes inside guardrails; one sender re-checks consent
3	Replies "Yes, I need someone to inspect my roof."	AI router + qualification; lead score calculated in SQL
4	Gets 3 real times	Google free/busy + business hours + database; AI never writes a time
5	Replies "2" and gets a confirmation	Live re-check → calendar event → database constraint → CRM deal
6	(Sales team) 🔥 NEW BOOKED LEAD on Slack	Name, problem, address, time and score

Then, on schedule: reminders 24 h and 2 h before the visit, no-show rebooking, quote follow-ups and a daily KPI report.

Engineering decisions

DECISION	WHY A CLIENT CARES
AI output is a validated decision, not an action; low confidence goes to a person	The bot can't book or promise anything on its own
Only times the system offered can be booked; live re-check + Postgres exclusion constraint + calendar rollback	No invented availability, no double bookings
One message sender that re-checks consent at send time	STOP is honoured everywhere, with an audit trail

Idempotent webhooks (phone, MessageSid, CallSid)	Twilio retries never double-text a customer
Retry queue re-runs from the failed node (1 / 5 / 30 min); health checks every 5 min	Outages recover on their own and the owner hears first
Business rules live in configuration, workflows are generated and deployed from code	A new client is configuration, not a rebuild

Results from testing

SCENARIO	RESULT
Web form lead	Saved in 224 ms; personalised first text
"Yes, I need someone to inspect my roof."	Urgency HIGH; 3 real times offered
"2"	Booked: appointment, deal, confirmation text, Slack alert
Same slot booked twice	Second attempt rejected, alternatives offered
"How much does a new roof cost?"	No price given; handed to a person; AI paused; Slack alert
Missed call	Text-back in seconds, once per call
Same Twilio message delivered twice	Processed once
"STOP"	Opted out; all follow-ups cancelled

> **John:** Yes, I need someone to inspect my roof.
> **RoofRadar:** (offers 3 free times from the calendar)
> **John:** 2
> **RoofRadar:** Your PrimeHome Roofing roof inspection is confirmed for Tuesday, Nov 10 at 9:00 AM...

Test setup: 34 executions on n8n 1.123 and PostgreSQL 16, with OpenAI, Twilio, Google Calendar and Slack answered by stand-ins that return each API's real response format, so every node, SQL statement and branch ran for real. Calendar booking and Slack alerts were also confirmed on the n8n Cloud deployment with live accounts.

Beyond roofing

Prompts, services, scoring and business hours are configuration, so the same core runs solar consultations (SolarFlow AI), property viewings (PropFlow AI), HVAC, plumbing, med spas and clinics.

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